Project INSPIRE Design Document

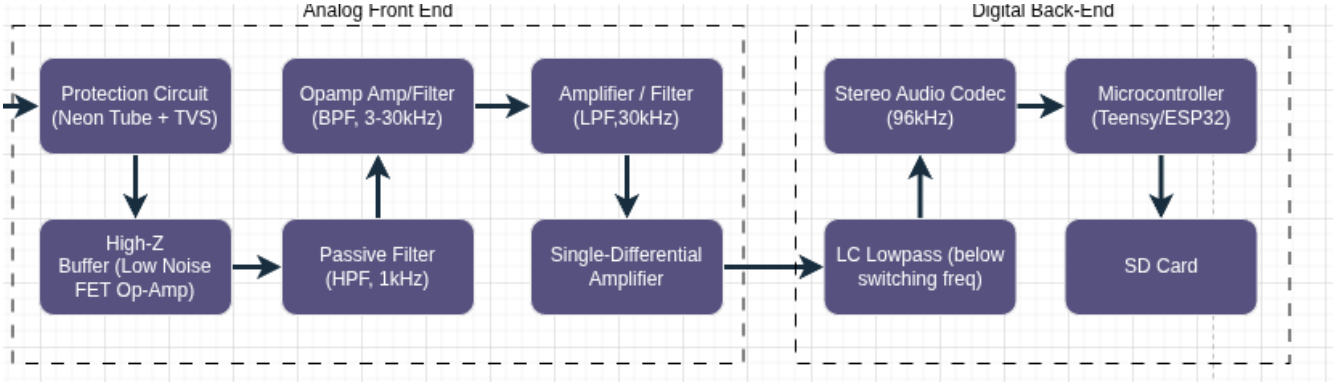
Project Part I: Antenna Filter & Protection Circuit

Revision 1.0: Presented 10/9/2026

1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:



The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

2) Sub-circuit Specifications

The Antenna Filter & Protection Circuit is one of the most important components of the entire signal chain, as its behavior will get transferred through the signal chain and any issues/nonlinearity will get amplified and interfere with the overall behavior of the system.

However, this circuit is essential, as the E-field whip will encounter a strong signal from the power line harmonics, which can damage the later amplifiers if we overvolt the amplifiers or ADC. Furthermore, the E-field whip, being a conducting metal rod, will give a static buildup and provides a risk for a high-voltage event. We do not want a direct strike to fry any components.

This means that a circuit will need three parts: A High-Voltage Bypass, a Transient-Voltage Supression (TVS) diode, and a shunt resistor in order to build a passive first-order High-Pass Filter (HPF).

- The high-voltage bypass will be accomplished by the use of a Gas Discharge Tube (GDT) which connects the antenna input to ground. This element does not conduct until an extreme voltage potential (typically ~70V) is provided across the element, which will ionize the gas inside. This creates a conductive pass with an extremely low resistance, which will short the circuit to a shunt ground. Ideally, this ionization also melts a small bit of metal inside that means that it keeps the voltage shorted.
    - While this does cut short any measurement, it is a sacrificial part that will prevent damage to other components or operators.
    - A GDT has not been selected or provided, so this needs to be one of the parts selected by the engineer. Look at the datasheet parameters for conduction voltage, effective resistance, and failure modes.

- For voltages higher then the operating rail of the design, but not high enough to trigger a GDT, a TVS diode can suffice. These parts act like a typical diode, but with their voltage threshold set to typical operating points. This means after a voltage difference is reached exceeding the specification, the resistance drops and it shunts to ground.
    - The voltage rails of the analog and digital supply are ±3.3V, which means a dual-rail 3.3V TVS diode network needs to be selected.
    - Make sure during selection to read the datasheet for the effective resistance, as this effects the filter circuit. (Parallel resistances drop)
- The passive filter will use the inherent capacitance of the antenna. Given an extremely small value capacitor (15-20pF), the filter resistor must be an extremely high value in order to ensure the cutoff frequency is the desired **1kHz**.
    - If you have taken Linear Circuits II, a first-order passive HPF should be simple. If you haven’t, research of their behavior is in order.

3) Expected Outputs

Due to lab restrictions, no high voltage prototyping can be done. As a result, this circuit design requires a writeup with three major parts:

- A selected model of Gas Discharge Tube, completed with parameters and a small paragraph justifying the part selection.
- A selected model of Transient Voltage Suppressor for the 3.3V dual-rail power supply, with parameters and paragraph for justification.
- A SPICE simulation of the RC lowpass filter given the 17.5pF mid-range capacitance, including resistor value.

Optional Extra (Stretch Goals):

- If all else is completed, see about importing a TVS diode model/including parameters in the simulation, and simulate the overall performance, including effective impedance and noise behavior.